

26AI5703- Data Engineering & Big Data Systems Lab

M.Tech AI&DS I

Exercise 4: PySpark

1. Install spark and pyspark
 - a. Run a spark shell and test the installation.
 - b. Run the wordcount program that you did using hadoop using pyspark.
 - c. Use the movielens dataset and find out for each movie, how are the ratings distributed.
2. Use the “friends_test” dataset. Col1 is ID, Col2 is name, Col 3 is Age, Col 4 is num of friends. Understand mapvalues function of RDD in spark and find the average number of friends for each unique age present in the dataset.
3. Use the “temp.csv” dataset. Column headers are present in the dataset. Understand filter operations and filter out only the “TMIN” values from the “desc” column. With the resultant data (RDD) find the following:
 - a. Minimum temperature (overall)
 - b. Minimum temperature for every ItemID
 - c. Minimum temperature for every StationID.
4. Use the same dataset, filter only “TMAX” column and find the maximum temperatures just like the ones mentioned above.